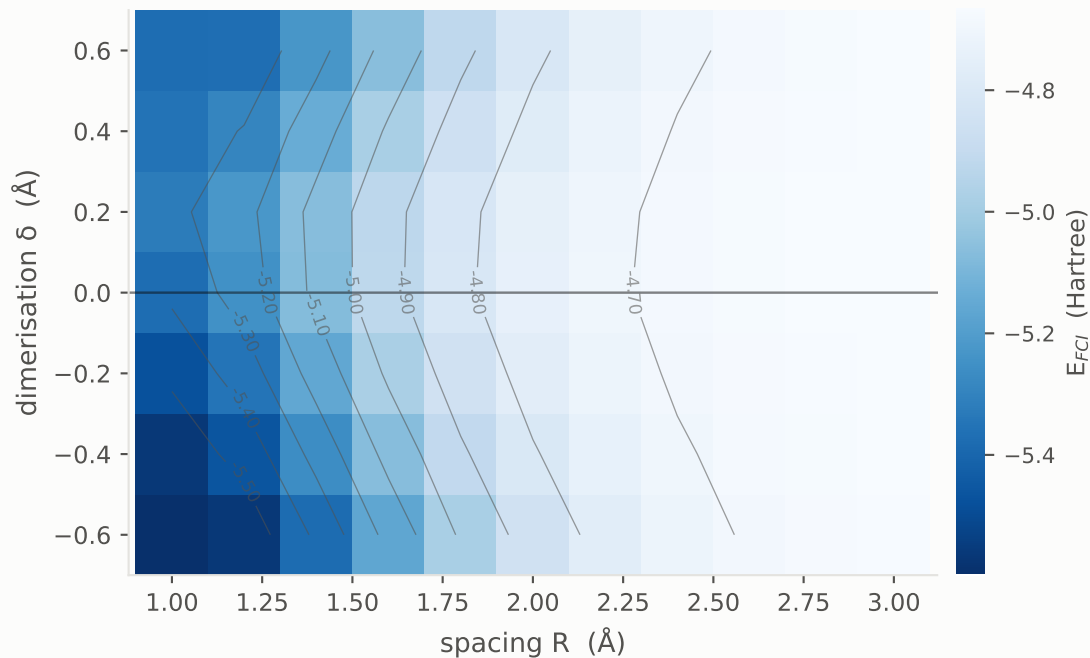


H_{10} (R, δ) surface, STO-3G — bonds alternate $R \pm \delta/2$

$\delta < 0$ gives five short bonds (five H_2); $\delta > 0$ gives four short bonds (four H_2 and two unpaired atoms). The chain has nine bonds, an odd number, so the two signs are different systems and the surface is not symmetric about $\delta = 0$. FCI is exact only for this basis, electron number and symmetry sector.

FCI energy surface



Per-geometry status

